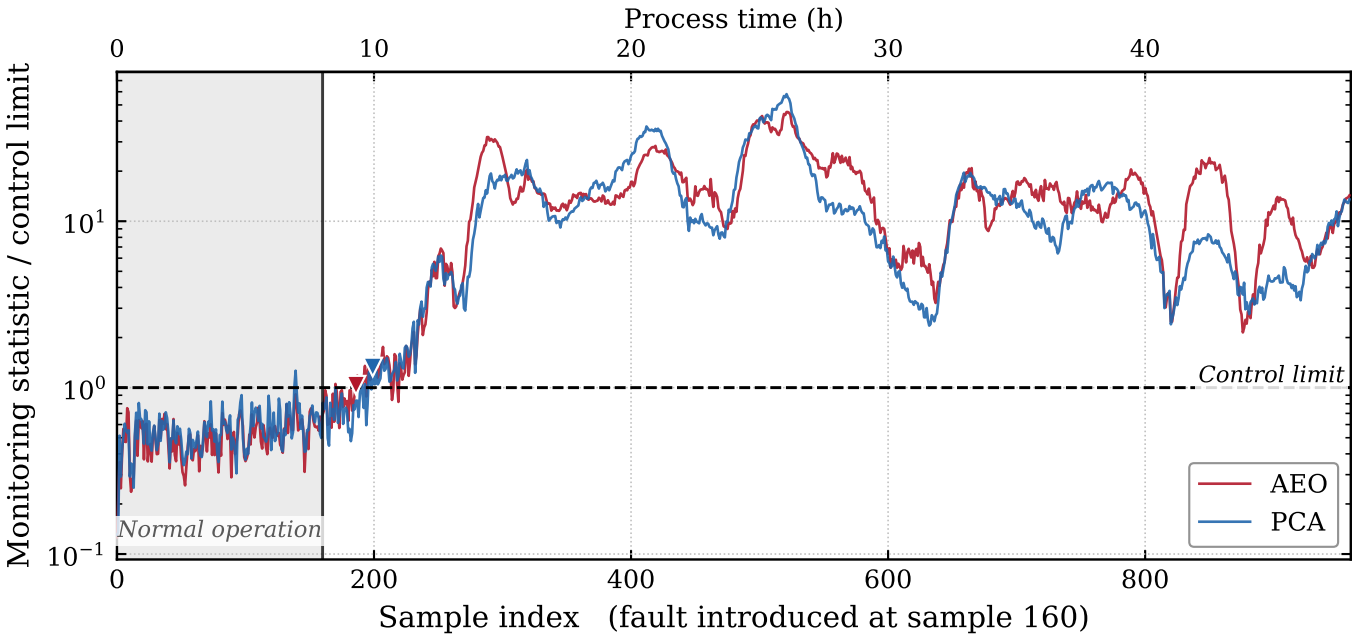
Fault Detection Performance for IDV(13)

Combined T^2 and SPE statistics — AEO vs. PCA at a common 1% false-alarm rate



Held-out performance (mean of 7 independent runs)

AEO	delay	43.4 samples (130 min)	MDR 0.059	detected in 7/7 runs
PCA	delay	40.6 samples (122 min)	MDR 0.052	detected in 7/7 runs

PCA detects 2.9 samples (9 min) earlier; PCA missed-detection rate lower by 13%